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CONGENITAL ANOMALIES AND HYDROCEPHALUS

Neural Tube Defects (NTD)

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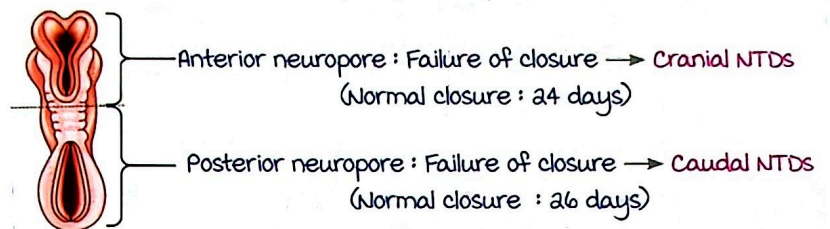
m/c congenital malformations in CNS (3.5- 4/1000 live births).

Risk in subsequent pregnancies : ax.

Risk factors :

1. Folic acid deficiency in mother.
2. maternal drugs : Anticonvulsant (**Valproate, Carbamazepine**).
3. Overt uncontrolled maternal diabetes.

Pathology :

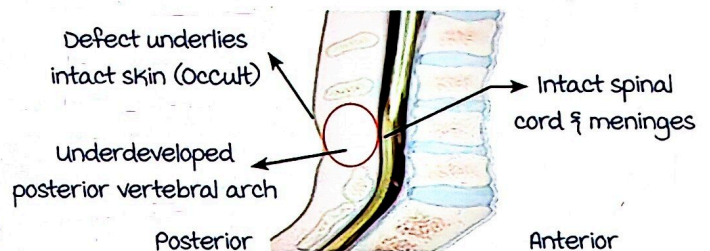


TYPES :

Caudal NTD's :

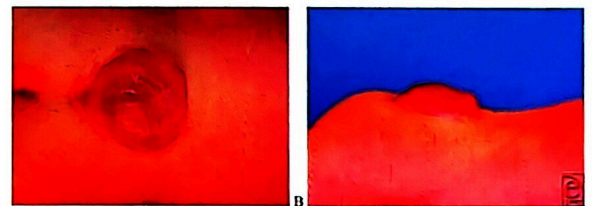
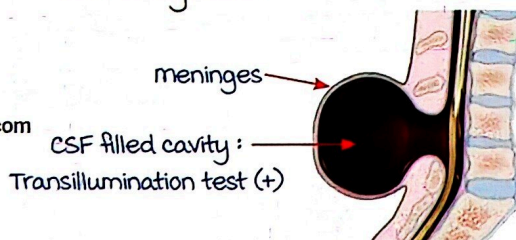
1. Spina bifida occulta :

- Asymptomatic condition.
- Clinical features : may present with swelling/tuft of hair over defect.



2. meningocele :

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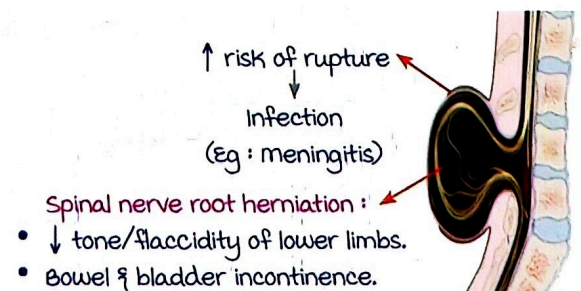


Spina bifida occulta

3. meningocele :

- "myelo" : Spinal cord structures.
- **most severe** caudal NTDs.
- Associated defects :
- **Arnold Chiari type II malformation (80%)**

↓
Hydrocephalus



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Neuronal Migration of Defects

- Defect of migration of neurons in CNS.

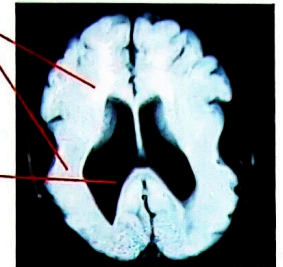
Lissencephaly :

Smooth appearance
of brain

Agyria (Absence of gyr)

↓ sed brain volume

Expanded ventricle



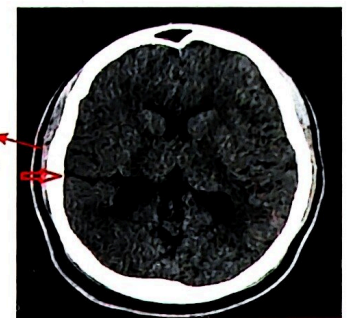
Schizencephaly :

- Clefts inside brain.
- Schizis : Splitting.

Porencephaly :

- Pore : Cystic spaces in brain.
- Cysts are lined by white matter.
- Can be secondary to cerebral infarctions : Intrauterine period/ perinatal (Birth asphyxia).

Cleft lined by
grey matter (+)



Schizencephaly

Agenesis of corpus callosum :

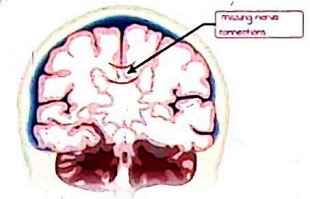
- Clinical features
 - microcephaly.
 - Low IQ.
 - Refractory seizures.

Aicardi syndrome :

- Associated with agenesis of corpus callosum.
- X-linked inheritance : Severe in females, lethal in males.
- Associations :
 - Eye manifestations.
 - Infantile spasms.
 - Vertebral defects (Eg : Hemivertebra).



Intact corpus callosum

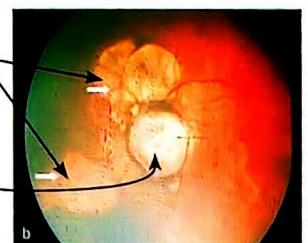


Agenesis of corpus callosum

Agenesis of corpus callosum

Characteristic
pits/lacunae

Coloboma



Eye manifestation

Holoprosencephaly :

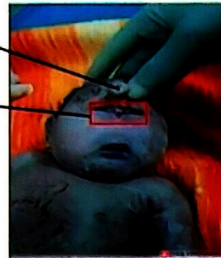
- Failure of division of prosencephalon → Single lobe brain.
- Embryonic forebrain → Cerebral lobes.
- Gene defect : Sonic hedgehog (SHH) gene.

Congenital Anomalies & Hydrocephalus

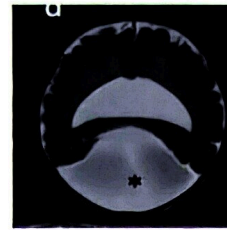
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- Associated genetic disorder : **Trisomy 13** (Patau's syndrome).
- Clinical features : midline facial abnormalities.

- Proboscis (Long, fused nose).
- Cyclops.
- Cleft lip/palate.



Associated facial abnormalities



Single lobe brain

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Hydrocephalus

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- Increased CSF volume in the brain (>50 mL).

Note :

- Normal volume of CSF → 150 mL (Adults).
- 50 mL (Infant).

CAUSES :

Obstructive

>>>>

1. Congenital anomalies

- Aqueductal stenosis** (m/c).
- Dandy Walker malformation.
- Arnold Chiari malformation.
- Vein of Galen (VOG) malformation.

2. Tumors : Posterior fossa tumors.

3. TORCH infections (m/c toxoplasmosis).

Non-obstructive

1. Increased production :

Choroid plexus papilloma.

2. Decreased absorption

(From subarachnoid space)

D/t

Blood :

Subarachnoid
hemorrhage

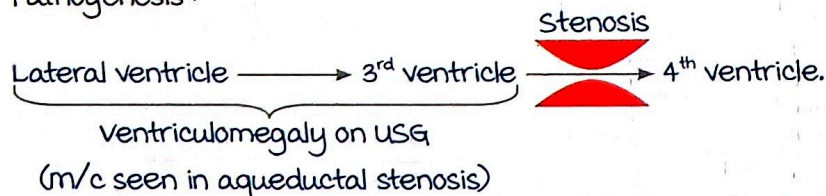
Pus :

meningitis

Aqueductal stenosis :

- m/c → Cause of congenital hydrocephalus
- Cause of fetal ventriculomegaly in USG

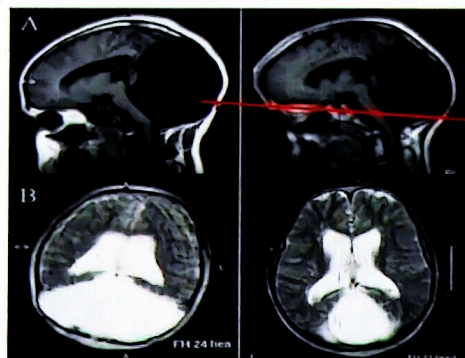
Pathogenesis :



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Nervous System

----- Active space ----- Dandy Walker malformation :



Dandy walker malformation

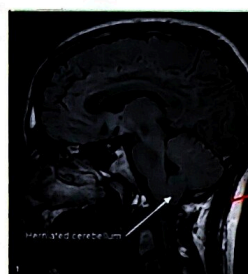
Cystic expansion of 4th ventricle

Dilatation of posterior fossa

1. Hydrocephalus.
2. Occipital widening.
3. Cerebellar hypoplasia.

Arnold chiari malformation :

Features :



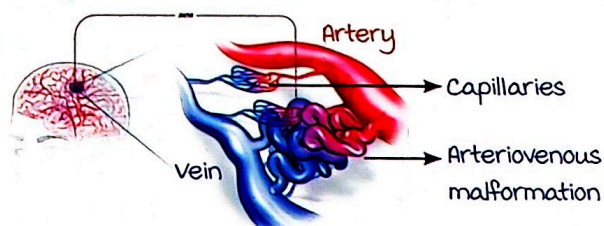
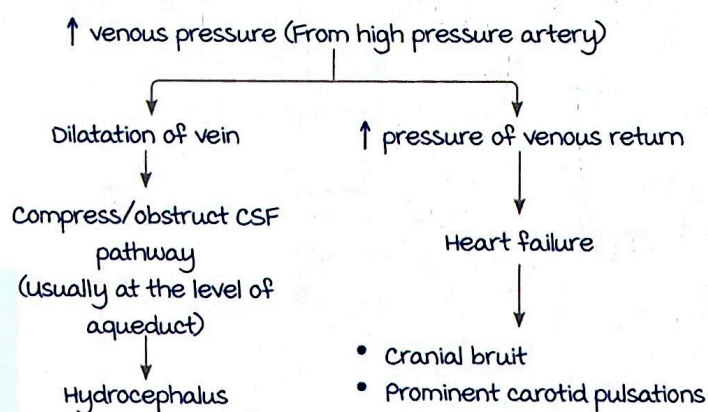
Foramen magnum

Herniation of cerebellar tonsil into
cervical canal of spinal cord

Types :

	Type I	Type II
Age group	Adults	Young children
Associations	Not associated with hydrocephalus	Hydrocephalus, meningomyelocele
Herniation of	Cerebellar tonsil	Cerebellar tonsil, brainstem structures, 4 th ventricle
Presentation	Neck pain, headache	-

vein of Galen malformation (VOG) :



Blood vessels in brain AVM



Dilated VOG

CLINICAL FEATURES

1. macrocephaly : $\uparrow$ HC $> +2$ SD.

2. $\uparrow$ ed intracranial tension (ICT) :

- Dilated scalp veins ($\downarrow$ draining).
- Bulging anterior fontanelles.
- Wide separation of skull sutures (>0.5 cm).
- **Sunset sign** : Visible upper sclera ($\downarrow$ / $\uparrow$ fixation in downward gaze).



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$\uparrow$ ICT $\rightarrow$ Dilated suprapineal recess $\rightarrow$ Compression of tectum in midbrain $\rightarrow$ Sunset sign.

INVESTIGATIONS :

Screening : Neurosonogram (USG at anterior fontanelle).

Gold standard/Definitive : **MRI brain**.

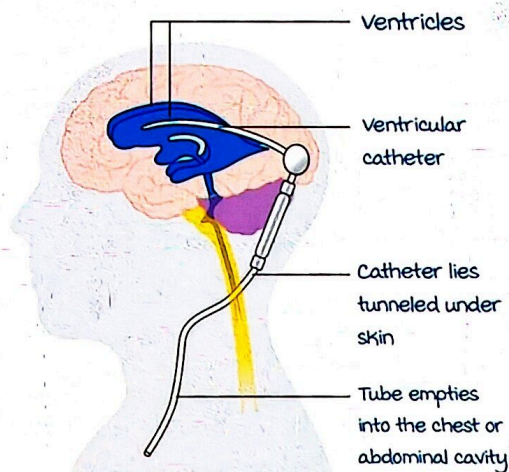
TREATMENT :

medical management
($\downarrow$ CSF production)

- **Acetazolamide**
(Diuretics, carbonic anhydrase inhibitor)

Surgical management

- V-P shunt (ventriculo-peritoneal)



- Complication : Shunt associated meningitis
 - Organism : **Staphylococcus epidermidis**
 - Diagnosis : CSF from **shunt tapping** $\rightarrow$ CSF analysis